

Cybersecurity Incident Report:

Network Attack Analysis

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is SYN flood attack. In this scenario, an attacker floods the target server with repeated 'SYN' packet requests, consuming system resources while leaving connections half-open by never responding with the final 'ACK'. As a result, legitimate users face connection timeout errors. The logs show that this is a network level denial of service (DoS) attack, called a SYN flood attack, that targets system resources. This event could be a malicious actor trying to intentionally flood the server with repeated SYN packets requests disrupting the system workflow.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. The handshake consists of three steps:

- 1.SYN(Synchronize): The visitor sends a 'SYN' packet to the server to initiate a connection request.
2. SYN-ACK (Synchronize-Acknowledge): The server responds with a 'SYN-ACK' packet to acknowledge the request and allocates memory resources to hold the connection state.
- 3.ACK (Acknowledge): The visitor responds with an 'ACK' packet, confirming the connection and establishing the connection.

When a malicious actor sends a large number of 'SYN' packets simultaneously without responding with the final 'ACK', connections are left in a half-open state. This rapidly consumes the server's available memory spaces and socket connection slots.

The network traffic logs indicate that the server is being flooded with repeated 'SYN' packets, leaving connections stuck in a half-open state. This consumes all the available server memory and socket connection slots. As a result, the server is unable to accept new 'SYN' requests from legitimate users causing it to drop connection attempts and respond with connection timeout errors or '[RST, ACK]' packets.